

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	11 September 2026
Team ID	Individual
Name	Mohammad Naheed Jahan
Project Name	Employee Raise Issue – Record Producer & Service Portal Integration
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses ServiceNow as its application platform. Employees report workplace or technical issues through the Employee Center by means of a Record Producer. UI Policies and dependent choices keep the information structured and accurate, and the submitted issue is saved in a custom ServiceNow table. Support users can manage the issue, assign it to the right group, update its status and priority, and follow its progress within the ServiceNow platform.

Architecture Flow

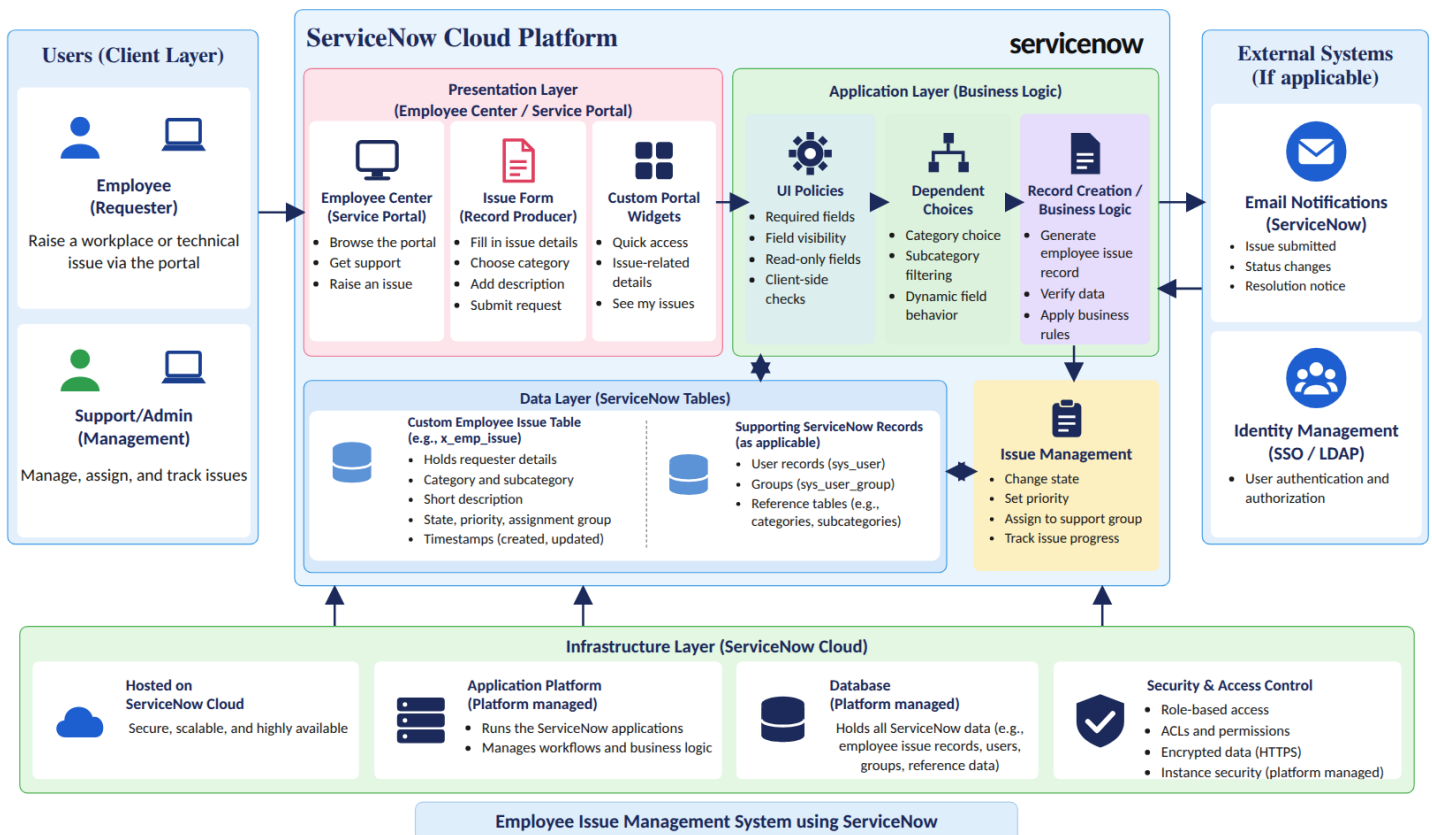


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	The interface employees use to open the Employee Center and submit workplace or technical issues.	ServiceNow Employee Center / Service Portal
2	Application Logic-1	Collects the employee's issue details and creates the matching issue record.	Record Producer, ServiceNow Forms & Variables
3	Application Logic-2	Manages field behavior and validation as the employee submits an issue.	ServiceNow UI Policies
4	Application Logic-3	Controls category and subcategory selection and shows the relevant choices based on the chosen category.	ServiceNow Dependent Choices / Choice Lists
5	Database	Holds employee issue information, including requester, category, subcategory, description, state, priority, and assignment details.	Custom ServiceNow Table / ServiceNow Tables
6	Cloud Database	Offers platform-managed storage for the application's issue records and related ServiceNow data.	ServiceNow Platform Data Store
7	File Storage	No separate file-storage component is needed for the core issue-reporting workflow.	Not applicable
8	External API-1	No external API is needed for software request processing.	Not applicable
9	External API-2	No external API is needed for software request processing.	Not applicable
10	Machine Learning Model	No machine-learning model is needed for the proposed workflow.	Not applicable
11	Infrastructure (Server / Cloud)	The application, portal, and business logic are hosted and run on the ServiceNow cloud platform.	ServiceNow Personal Developer Instance / Enterprise Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is needed. The solution is configured within the ServiceNow platform.	ServiceNow platform configuration
2	Security	Access to employee issue records and administrative functions is governed by ServiceNow authentication, roles, permissions, and access controls.	ServiceNow roles, ACLs, authentication and authorization
3	Scalable Architecture	The solution is platform-based and can support more software catalog items, users, approval rules, and fulfillment tasks as request volume grows.	ServiceNow platform and Flow Designer
4	Availability	The application runs on the ServiceNow platform, so employees and support teams can reach the configured issue-management system through the ServiceNow instance.	ServiceNow instance/Employee Center
5	Performance	The solution cuts down manual issue-reporting effort through a structured form, controlled data entry, automated record creation, and organized issue tracking.	Record Producer, UI Policies & ServiceNow Tables